AN IMPURITY DESCENDS TO SOME STAGE OF A DIRECTED LIMIT OF AFFINE SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a base change verification of the relative assassin condition followed by a limit and quasi-compactness argument producing a stage where the closure condition already holds

Editorial note. The label lemma-impure-limit names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Impurities above a point as witnesses of impurity, and the base change of the relative assassin of a module
- The reduced induced structure on the closure of a point, and its behaviour under a directed limit of the base
- Quasi-compactness of the fibres of a morphism of finite type, and the criterion that a directed limit of quasi-compact spaces is nonempty exactly when some stage is

Source reference. Original reference 05FW: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 260–260. Prerequisite pointer invoked at proof line(s) 320; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CV2: sources/stacks-chow/tags.txt, source lines 16122–16122. Prerequisite pointer invoked at proof line(s) 333; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CUK: sources/stacks-chow/tags.txt, source lines 16105–16105. Prerequisite pointer invoked at proof line(s) 344; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 086W: sources/stacks-chow/tags.txt, source lines 10406–10406. Prerequisite pointer invoked at proof line(s) 357; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-flat.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In Situation 05FW. Let $(g : T \to S, t' \rightsquigarrow t, \xi)$ be an impurity of $\mathcal{F}$ above y. Assume $T = \lim_{i \in I} T_i$ is a directed limit of affine schemes over Y. Then for some i the triple $(T_i \to Y, t'_i \rightsquigarrow t_i, \xi_i)$ is an impurity of $\mathcal{F}$ above y.*

Proof. The notation in the statement means this: Let $p_i : T \to T_i$ be the projection morphisms, let $t_i = p_i(t)$ and $t'_i = p_i(t')$. Finally $\xi_i \in |X_{T_i}|$ is the image of ξ. By Divisors on Spaces, Lemma 0CV2 we have $\xi_i \in \mathrm{Ass}_{X_{T_i}/T_i}(\mathcal{F}_{T_i})$. Thus the only point is to show that $t_i \notin f_{T_i}(\overline{\{\xi_i\}})$ for some i.

Let $Z_i \subset X_{T_i}$ be the reduced induced scheme structure on $\overline{\{\xi_i\}} \subset |X_{T_i}|$ and let $Z \subset X_T$ be the reduced induced scheme structure on $\overline{\{\xi\}} \subset |X_T|$. Then $Z = \lim Z_i$ by Limits of Spaces, Lemma 0CUK (the lemma applies because each X_{T_i} is decent). Choose a field k and a morphism $\mathrm{Spec}(k) \to T$ whose image is t. Then

$$\emptyset = Z \times_T \mathrm{Spec}(k) = (\lim Z_i) \times_{(\lim T_i)} \mathrm{Spec}(k) = \lim Z_i \times_{T_i} \mathrm{Spec}(k)$$

because limits commute with fibred products (limits commute with limits). Each $Z_i \times_{T_i} \mathrm{Spec}(k)$ is quasi-compact because $X_{T_i} \to T_i$ is of finite type and hence $Z_i \to T_i$ is of finite type. Hence $Z_i \times_{T_i} \mathrm{Spec}(k)$ is empty for some i by Limits of Spaces, Lemma 086W. Since the image of the composition $\mathrm{Spec}(k) \to T \to T_i$ is t_i we obtain what we want. □